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‘WHAT-IF’ ANALYSIS GUIDELINE

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1.0 PURPOSE

This Guideline is intended as guidance to Idd El Shargi personnel in how to execute ‘What-If’ analysis technique in line with the Idd El Shargi Procedure for Process Hazard review, required in IS-HSE-PRC-308.

The General objectives of the Idd El Shargi Risk Management Guidelines (RMG), within the Idd El Shargi RM process are to:

- Allow a clearer definition of requirements with regard to the Idd El Shargi RM program and related Policies and Procedures.
- Allow specific topics to be addressed in detail relevant to the application of the RM element, including examples where relevant, and
- Act as a basis for further information, understanding and training.

Note: These Documents should be considered as Guidelines only and not mandatory unless otherwise stated, the use of the words, ‘shall’ and ‘should’ will indicate a mandatory requirement. For further clarification on content and intent contact Idd El Shargi HSE Risk Engineer.

The purpose of this Guideline is to provide guidance on when and how to apply CMEs when calculating Consequences associated with an Event Scenario; and to ensure that, when evaluating Risk, a consistent approach to the application of Critical Mitigation Elements is applied.

2.0 SCOPE AND APPLICABILITY

The Guideline applies to all personnel within Idd El Shargi plants and facilities, who are involved in Risk Management, and applies to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any Idd El Shargi facilities and/or locations.

This Guideline covers all systems containing hydrocarbons or other hazardous fluids, utilities, safety systems, Greenfield and brownfield projects, campaign activities, and MOC related activities which may pose a risk to personnel, the environment, property damage and loss of revenue. This Guideline only addresses those aspects which are unique to the ‘What-If’ analysis technique. It does not cover the follow-up activities.

3.0 GUIDELINE

3.1 GENERAL

The ‘What-If’ analysis technique is a less systematic, multidisciplinary team oriented analytical technique, which relies on the experience of the participants. The ‘What-If’ analysis technique has been developed as an efficient method for providing highly effective risk identification and evaluation when it can be demonstrated that circumstances do not warrant the rigor methodology (i.e., HAZOP). HAZOP examines the plant line by line, vessel by vessel, etc. ‘What-If’ analysis on the other hand, is a systems-oriented technique which examines complete systems or subsystems. The technique is efficient because it generally avoids lengthy discussion of areas where the hazards are well understood or where prior analysis has shown no hazards are known to exist. Its effectiveness in identifying hazards comes from asking questions in a variety of important areas, to help ensure complete

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coverage of all the various types of failures or errors which are likely to result in a hazard within the system being examined.

In Idd El Shargi, the ‘What-If’ process is a team-based brainstorming technique using scenario-based events, and sometimes it requires a deviation type approach. It can be employed in addition to or instead of a full HAZOP where the degree, nature and type of change so permits. ‘What-If’ analysis technique could be employed for issues relating to the loss of a critical component, for example, where a routine inspection or test reveals the failure of an ESDV to close on demand. A typical example for a ‘What If’ would be a singular change to a trip function on a critical system, e.g. a hydrocarbon system.

The key objective of any ‘What-If’ analysis in Idd El Shargi is to identify hazards, risk assess, determine additional controls for risk reduction and highlight recommendations for further study/ review. The objective of this Guideline is to give guidance to Idd El Shargi personnel in how to execute such technique for the following categories of questions:

- Material Problems.
- External Effects or Influences.
- Operating Errors and Other Human Factors.
- Analytical or Sampling Errors.
- Variance to Standard or Procedure requirements (e.g., isolations, detection coverage, etc.).
- Safety critical equipment maintenance/ inspection deferrals.
- Equipment/ Instrumentation Malfunction.
- Process Upsets.
- Utility Failures.
- Integrity Failure (i.e., potential of LOPC).
- Emergency Operations.
- Campaign activities.
- Non-routine, transient, high risk potential activities, etc.

3.2 “WHAT IF” ANALYSIS REVIEW TEAM

3.2.1 TEAM STRUCTURE

In Idd El Shargi, ‘What-If’ analysis should be led by an independent ‘What-If’ analysis facilitator.

The ‘What-If’ analysis team shall consist of:

- Independent Facilitator may be supported by a scribe.
- Project/ Change (MOC) Proponent.
- Engineering Representative.
- HSE and/ or Technical Safety Engineering Representative.
- Operations Representative.
- Maintenance Representative.

Team members who may be called upon as required include:

- Asset Integrity Representative.
- Construction Representative.

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- Marine Specialists.
- Drilling/ Well Service Engineers.
- Specialist Engineering Representatives (Technical Authorities).
- Consultant/Third Party Specialists.
- Vendors Representatives.
- Others as required.

It is essential that the team composition be appropriate for the facility or design modification being studied.

In general, the role of team members is to creatively share their collective operating experiences and process knowledge to:

- Identify credible scenarios and accident sequences for discussion.
- Identify and challenge effectiveness of existing safeguards.
- Qualitatively evaluate risk of each identified scenario.
- Make appropriate recommendations aimed at reducing risk.

All persons attending a 'What-If' analysis should ideally be in attendance for its full duration, without substitution. All 'What-If' analysis team members should ideally be formally trained in the Idd El Shargi 'What-If' analysis process.

3.2.2 "WHAT-IF" ANALYSIS FACILITATOR

The 'What-If' analysis facilitator should be experienced with relevant knowledge in the oil and gas industry, but a generalist rather than an 'expert'. He should be experienced, qualified and certified in facilitating 'What-If' analysis studies of a similar level of complexity.

Responsibility for the conduct of the 'What-If' analysis lies with the 'What-If' analysis facilitator. The "What-if" analysis facilitator has the responsibility to ensure that the 'What-If' analysis study proceeds in an effective and efficient manner, maintaining the essential flow of the study, its robustness and in ensuring the active involvement and participation of all members of the "What-if" team.

The 'What-If' analysis is recorded on a risk assessment worksheet software (e.g., PHA Pro), very similar to that used for HAZOP recording. The 'What-If' Analysis Facilitator, with the assistance of the scribe, is responsible for 'What-If' analysis record sheets, for agreement by and acceptance of the team and in preparation of the 'What-If' analysis report.

The 'What-If' Analysis Facilitator does not have responsibility for implementation or close out of action items arising from the 'What-If' analysis.

3.3 METHODOLOGY

3.3.1 REVIEW DEFINITIONS

A significant amount of preparatory work is required both by the Project/ Change proponent and by the "What-if" facilitator prior to each session. This work typically comprises:

- Collecting the relevant data, e.g., FTIs, work pack, method statement.
- Validation of information.
- Planning the sequence of the "What-if" analysis, etc.

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Once the preparations are completed and the ‘What-If’ analysis team assembles, the team should spend a brief period of time to discuss the Purpose, Scope, and Objectives of the analysis. The initial establishment of Purpose, Scope and Objectives, is very important and should be precisely set down so that it will be clear at the time of the study and in the future, of the basic intent. There is a specific sheet to be completed in the Administration section of the PHA Pro ‘What-If’ analysis template for the inclusion of Purpose, Scope, and Objectives, and defined as follows:

- The purpose defines the driver behind the analysis e.g., procedural requirements-a yet to be built project, or facility changes; Incident outcome requirement, e.g., using a ‘What-If’ analysis to review a process system involved in the incident in question. The purpose should take the form of a succinct statement clearly identifying why the PHR analysis is being undertaken.
- The scope of the review is the boundaries of the physical unit, and also the range of events and variables considered. It should also highlight any areas that are not specifically covered.
- The objectives should be distinct statements which expand upon the defined purpose.

The project or change (MOC) proponent should brief the team about important information, including:

- The technical reason or basis for the change.
- The expected impact the change might have on safety, health, environment, property damage, and potential loss of revenue.
- The need to change or modify operating Procedures.
- The intended duration of the change and if possible, an estimate of how long its start-up is likely to take, etc.

At the end of PHR sessions, the scribe also needs to record and highlight the summary of the reviews, e.g., any PEC issues, potential D/ E level risks, transient risk/ time-at-risk issues, etc.

3.3.2 “WHAT-IF” ANALYSIS REVIEW DOCUMENTS

For MOC type project and Ad-Hoc reviews, Project or Change (MOC) proponents will be responsible for assembling the relevant data to meet the requirements of the “What-if” analysis Facilitator. All ‘What-If’ supporting Documents should be issued to all participants for their review and preparation for the ‘What-If’ prior to the start of the ‘What-If’ analysis study.

The following Documents may be required by the ‘What-If’ analysis team:

- Process FTIs (e.g., PFD’s, P&ID’s for Process & Utilities, Heat and Mass balances, Process Design Basis, etc.).
- Electrical and Instrumentation FTIs (e.g., Cause and Effect Diagrams, Single Line Diagram, Process Operation & Control Philosophy, etc.).
- Drilling and Well Service FTIs (e.g., Well Schematics).

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- Operations FTIs (e.g., Operations and Maintenance Manual, Material Safety Data Sheets, etc.).
- General Arrangements drawing (Plot Plans / General Equipment Arrangements & Elevation Drawings, etc.).
- Safety FTIs (Safety Layout drawings, F&G drawings, Hazardous Area Classification drawings, etc.).
- MOC supporting Documents (e.g., Method Statements, Workpack Documents, etc.).
- Idd El Shargi HSE Risk Matrix and Response Chart.
- CME/ KME Cheat Sheet.

Additional Documents may be called for during the ‘What-If’ analysis study to amplify or verify information requested by the ‘What-If’ team.

3.3.3 REVIEW SESSIONS

3.3.3.1 SYSTEMS & SUB-SYSTEMS

The division of the project/ facility under ‘What-If’ analysis study should be delineated into appropriately sized segments termed System/Sub-systems. A system may contain general information of facility or any process subject under review. A system may contain single or multiple subsystems. A Sub-system may contain process lines and/ or equipment, general information of equipment, e.g., Equipment ID, Design Conditions and Parameter.

Example for System and Sub-system is provided below on Table 1.

TABLE 1. SYSTEM & SUB-SYSTEM EXAMPLE

System	Sub-system
PS-1D platform	First stage KO Drum (V-4101)
	First stage separator (V-4102)
	First Stage Gas Compressor (K-4110 A)
	Second Stage Gas Compressor (K-4110 B)
PS-1 Control System	DCS – Process Control
	ESD System
	Fire & Gas Detection

3.3.3.2 CONDITIONING FACTORS

Any conditions affecting equipment and piping (i.e. General condition of equipment / piping, Deferrals, Documented Failures / Problems, Temporary repairs / wraps, NRO's (non-routine operations), LTI's (long term inhibits), Outstanding actions from related PHA's, PSSR's and Punch lists), which may impair the robustness of an Intrinsic Control should be identified as a Conditioning Factor and the related CME disregarded as an appropriate control or reduced in status to a KME depending on the severity of the conditioning factor.

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Given the brownfield nature of Idd El Shargi facilities, the inclusion of Conditioning Factors in the analysis is essential in order to establish the current effectiveness of the CME and/or KME impacted. A qualified judgment must be made on any Conditioning Factors that affect a CME's effectiveness on a system and if the outcome is critical, an action item should be recorded, e.g., Long Term Isolation (LTI) on an ESDV will mean it is no longer considered effective in preventing or mitigating an Event Scenario.

Conditioning Factor types are provided in Appendix 1 of this Guideline Document.

3.3.3.3 WHAT-IF SCENARIOS

All credible what-if questions shall be identified. For all ‘What-If’s identified against a specific sub-system, all categories of consequence shall be addressed, e.g. Personnel, Environment, PD-LOR (Property Damage Loss of Revenue) and Impacts outside a facility and duly documented. Idd El Shargi templates require all 6 consequence categories below, to be covered for each scenario if applicable:

- People
 - People at the Facility (PAF).
 - People Outside the Facility (POF).
- Property Damage/ Loss of Revenue (PD/ LOR).
 - Non-owned Property Damage (NOP).
- Environmental
 - Environmental Remediation (ER).
 - Environmental Wildlife Habitat (EWH).

If a particular category is not applicable to a study at all, then it can be covered once in the PHR worksheet with a general note, see below statements as example:

- No surrounding non-Idd El Shargi owned properties or businesses; therefore ‘NOP’ category will not be assessed further in the ‘What-If’.
- No surrounding protected species or habitats; therefore ‘EWH’ category will not be assessed further in the ‘What-If’.

‘What-If’ analysis questions shall be defined to include the ‘why’, the ‘how’, and where relevant, the ‘when’. All causes must be verified to have valid initiating events, i.e. they need to have a unique cause consequence relationship. The Guidelines for identifying causes are as follows:

- Team brainstorm.
- Address all types of causes e.g., human error, equipment failure, or external events.
- Record all causes first prior to risk ranking.
- Group equipment failures as appropriate.
- Be specific i.e., blockage does not define method of prevention and operational issue, so blockage due to debris is more specific.
- Verify all causes are valid initiating events.

The three major categories of causes, in decreasing probability are:

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- a. human error,
- b. equipment failure,
- c. external events.

The Team should consider the levels of definition of the causes to ensure adequate detail is provided. There are 3 levels of definition to be considered during ‘What-If’ analysis they are:

- Level 1 (immediate cause) e.g., Pump fails off.
- Level 2 (basic cause) e.g., Pump fails off due to Mechanical Failure.
- Level 3 (Enabling event or conditioning factor) e.g., Pump fails off due to mechanical failure due to lack of PM.

The Level 2 (basic cause) is to be regarded as an appropriate level of cause.

Causes must be credible; a non-credible event would be one which is, possible but highly improbable. Causes should also not involve double jeopardy events, i.e. they should not involve two or more unrelated events or conditions that would have to occur concurrently in order to initiate the sequence of events that leads to adverse consequences. If non-credible causes are considered by the team this may lead to expensive and unnecessary modifications to the facility being implemented.

Details of Process Event Scenario development are provided in IS-HSE-GDL-310, the ‘Developing Process Event Scenarios’ Guideline.

3.3.3.4 CONSEQUENCE ESTIMATION

Consequence - shall be analysed before Probability, using the unmitigated, ‘worst-case’ credible consequence (i.e., reasonable consequences, NOT absolute worst case), for the consequence category in question (e.g., Personnel, Environment, Property Damage and Loss of Revenue, and External impacts).

Consequence should be addressed in the context of Hazard Scenario and Risk category (e.g., Personnel), as a series of events, and to the point of ‘Impact Scenario’. For a consequence category of PD/LOR, consequence should be quantified in terms of financial losses over a period of time.

Consequence Rule Sets for Major Hydrocarbon Loss of Containment Scenario:

- Loss of containment due to over pressurization case for equipment supported by the Mechanical Integrity program.

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TABLE 2. OVER PRESSURISATION CONSEQUENCE RULE SETS

Deviation	Degree of Deviation	Consequence
More Pressure	Up to and including system/ equipment test pressure	Minor flange leaks
	Up to 50% above test pressure	Leaks from flange or instrument fittings
	>50% above test pressure	Line failure/ vessel rupture and large release

- H₂S release impact to personnel at facility.

TABLE 3. H₂S RELEASE IMPACT TO PERSONNEL RULE SETS

H ₂ S Consequence Category	1	2	3	4	5
	Consequence Level				
Human injury at Facility	≤ 500 ppm	500 ppm ≤ x ≤ 700 ppm	> 700 ppm H ₂ S risk of PII	> 700 ppm H ₂ S risk of PII	> 700 ppm H ₂ S risk of PII

For all manned areas, the default consequence level for PAF caused by a Hydrocarbon Loss of Containment leading to toxic release and/ or fire event from a 100mm dia. orifice or greater shall be taken as 2 or 3 PII (i.e. consequence level 4), as a minimum.

3.3.3.5 INTRINSIC CONTROLS

Intrinsic Controls are controls that can be assumed are/ will be in place. In Idd El Shargi there are two types of intrinsic controls that may be considered, they are: CME (Critical Mitigation Element), and KME (Key Mitigation Element), described as follows:

- Critical Mitigation Element (CME) - An independent and reliable device or system that is capable of preventing or mitigating an undesired Event Scenario. Failure of one CME shall not prevent another CME from performing its function. CME shall be maintained to the reliability level assumed in the risk analysis. CME's are a subset of Key Mitigation Elements (KMEs).
- Key Mitigation Element (KME) - Equipment, design features, specifications, inspection and maintenance routines, operating parameters and other HSE Risk Mitigation elements considered in a risk analysis that are necessary to achieve and maintain the reliability levels assumed in the risk analysis.

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The use of Administrative type / Human intervention controls (KMEs), are generally not considered 'robust' controls and careful consideration and consultation with Idd El Shargi HSE- Risk Function and Idd El Shargi Risk Engineering must be undertaken if they are utilized to achieve Risk reduction from the C, D, E levels.

Details of Critical Mitigation Element (CME) and Key Mitigation Element (KME) are provided in IS-HSE-GDL-309, the 'CME Methodology' Guideline.

3.3.3.6 INITIAL RISK

In the Initial Risk assessment, 'Intrinsic Controls' are identified to determine the probability level using Idd El Shargi HSE Risk Matrix. If Initial Risk is not at a tolerable level, Additional Controls should be recommended to reduce the risk to ALARP.

For Initial Risk, levels of D, E, where the team cannot identify any Additional Controls to reduce the risk, the Project/ MOC proponent should report this immediately to Idd El Shargi Risk Engineering Team for further evaluation.

3.3.3.7 POTENTIALLY ELEVATED CONSEQUENCE (PEC)

Some event scenarios have much higher consequences than the threshold value for a level 5 consequence. The study team shall include PEC Codes where a consequence level of 5 is assigned and when the:

- PEC-1; Potential consequence to Personnel exceeds 14 Permanent Incapacitating Injuries onsite, offsite, or cumulatively.
- PEC-2; Potential consequence in combined Property Damage (PD) and Loss of Revenue (LOR) exceeds \$1 billion.
- PEC N/A; If a PEC does not apply when a consequence level of 5 is determined.

3.3.3.8 HSE CRITICAL

In the event that a Consequence Level is determined to be a 4 or 5 then the relevant intrinsic and/ or additional controls shall be considered as HSE Critical and identified as such by input of a 'Y' in the appropriate HSE Critical Equipment column of the 'What-If' Worksheet, see Appendix 2 below. The related CMEs and KMEs will also be considered to be HSE Critical.

3.3.3.9 ADDITIONAL CONTROLS / RECOMMENDATIONS

Additional Controls/ Recommendations are Facility or change-specific controls usually identified in the 'What-If' analysis in order to reduce risk from the levels achieved by the intrinsic controls to ALARP.

Additional Controls shall be categorized accordingly (i.e., CME - Eliminate, CME - Prevent, CME - Detect, CME - Mitigate, Respond, Recover, and KME), and these categories shall be carefully considered for the resultant Risk evaluation. Emphasis shall be given to the principles of Inherent safety (e.g., Elimination of the Hazard and the approach of Minimise / Substitute / Moderate / Simplify), where feasible, and Prevention, e.g., reduction of likelihood of the event/impact scenario.

Moreover, as stated above in "Intrinsic Controls", use of Administrative type / Human intervention controls (KMEs), are generally, not considered 'robust' controls and careful

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consideration and consultation with Idd El Shargi Risk Engineering must be undertaken if they are utilized to achieve Risk reduction from the C, D, E levels.

In addition to specific ‘Additional controls’ the ‘What-If’ team should include ‘recommendations’. Note that no credit shall be taken for ‘Recommendations’ when determining Residual Risk; only ‘Additional Controls’ with a clear indication that those will be implemented shall be considered in Residual Risk determination.

It is important when determining required actions (Additional controls and recommendations), that indecisive and vague wording is avoided. The item should be recorded such that it can be acted upon by the person designated to execute it. The actions proposed should be feasible and practicable and not totally unrealistic in the context of Idd El Shargi risk. The actions may be challenged by the change owner or other responsible parties and this may result in the need for some form of cost benefit analysis.

3.3.3.10 RESIDUAL RISK

The ‘Residual Risk’ should be determined by the team by means of the modified probability or consequence levels based on the identified Additional Controls, which determine the potentially changed Risk position in the Idd El Shargi Risk Matrix. In the event that no Additional Control or recommendation is identified then the Residual Risk shall be recorded on the worksheet at the same value as the Initial Risk.

3.3.3.11 ‘WHAT-IF’ ANALYSIS FLOWCHART

The flowchart in Figure 1. below shows the assessment process to be followed.

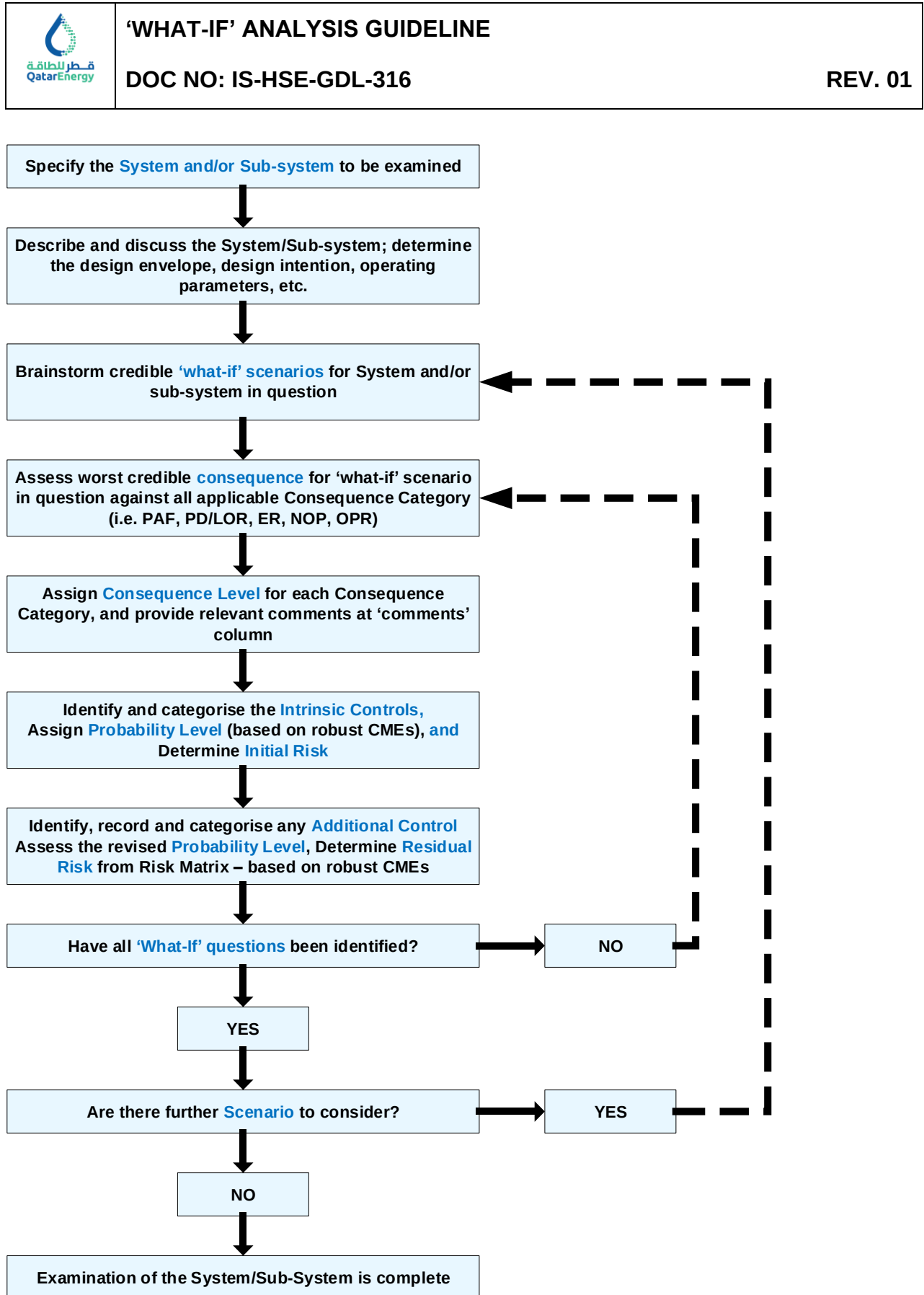


FIGURE 1. 'WHAT IF' ANALYSIS FLOWCHART

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3.3.4 REPORTING

The ‘What-If’ analysis report should utilise Standard Idd El Shargi ‘What-If’ analysis template output report from software (i.e., PHA Pro). RM administrator to issue ‘revision A’ (e.g., rev. A1, A2, etc. as required), report to all team members for their review and comments. The report content shall be agreed by the team members. Once comments from the team members have been incorporated, the RM administrator should issue ‘revision B’ (e.g. rev.B1, B2, etc. as required), report for approvals and actions completion by action owners. A copy of all reports shall be maintained by the RM administrator for record and distribution purposes.

The following should be reported to Idd El Shargi Risk Engineering for further evaluation and subsequent reporting as required by IS-HSE-PRC-301:

- Residual Risks at the D / E Level.
- For Initial Risk levels of D, E, where the team cannot identify any Additional Controls to reduce the Risk. (Note: A "Recommendation" shall be made in the ‘What If’ report to require further, more detailed review, with the Residual Risk level reported as the initial. This ensures the Risk level will be highlighted in the ‘What If’ Study / Summary report.
- Where the residual Consequence level is 5, for any category of Risk.
- Any scenario with consequence of PEC 1 and PEC 2.

The reports should be issued in line with the requirements of IS-HSE-PRC-301 the Risk Evaluation and Response Procedure.

The Standard ‘What-If’ analysis report should contain:

- A worksheet report (automatically generated by the PHA-Pro software).
- Supporting Documents (e.g., mark-up of P&IDs, plot plans, workpack Documents, etc.).

A completed review report can be used to demonstrate to interested parties that a ‘What-If’ analysis has been accomplished and all possible actions have been examined and or implemented to eliminate major hazards.

All the drawings, reference data, evaluations, record sheets, action sheets etc., used or developed during the course of the ‘What-If’ analysis should be filed for future reference.

On completion of the ‘What-If’ analysis, the RM Administrator will compile a report, describing the scope of the work and containing copies of the main Documents and drawings used. A typical report will contain a brief description of project or change proposals including Purpose, Scope, and Objective statements, list of team members, dates of meetings, list of documentation being used as basis of review, actions summary and recommendations. Other principal Documents may be presented as an Appendix. The final report is circulated and agreed to by all team members.

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4.0 REFERENCES TO OTHER DOCUMENTS

TABLE 4. REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
Risk Evaluation and Response Procedure	IS-HSE-PRC-301	Internal	Upwards
Process Hazard Review Procedure	IS-HSE-PRC-308	Internal	Upwards
Critical Mitigation Element Methodology	IS-HSE-GDL-309	Internal	Sideways
Developing Process Event Scenario	IS-HSE-GDL-310	Internal	Sideways
Petroleum and natural gas industries – Offshore production installations Guidelines on tools and techniques for hazard identification and risk assessment	ISO 17776: 2000	External	N/A

5.0 DEFINITIONS

TABLE 5. KEY TERMS DEFINED

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Term	Definition
CME	Critical Mitigation Element, An independent and reliable device or system that is capable of: 1) preventing initiation of an Event Scenario, 2) preventing loss of Hazardous Material containment, or 3) substantially mitigating one of an Event Scenario's Consequences. Failure of one CME shall not prevent another CME from performing its function. CME shall be maintained to the reliability level assumed in the risk analysis. A CME reduces an Event Scenario's Probability by one level on the Risk Matrix.
HAZOP	Hazard and Operability Study, A formal team-based, structured, hazard identification and risk assessment method that applies a detailed deviation, guideword, parameter, cause, consequence, and controls approach. Its main objective being to validate a robust design in terms of Hazards And Operability - it is not intended as a design review.
KME	Key Mitigation Element, HSE Risk mitigation elements (equipment, design features, specifications, inspection and maintenance routines, and operating parameters) that are necessary to achieve and maintain the reliability level of Critical Mitigation Elements (CMEs), and Risk Mitigation Measures (RMM), or risk mitigations that are not otherwise claimed as CMEs / RMMs in a risk analysis.
NRO	Non-Routine Operations: Non-Routine Operating Instructions are formal documents describing and sanctioning departures from normal operating practice. Such temporary operations are managed via the MOC process.
PEC	Potential Elevated Consequences, Consequences of an Event Scenario that exceed 14 Permanent Incapacitating Injuries or \$1 billion of combined PD/LOR and thereby substantially exceed the threshold for Consequence Level 5.
PII	Permanent Incapacitating Injury, A fatality or physical injury or illness that permanently renders a person incapable of performing the essential duties of any of the occupations for which a person is qualified by education, training and experience
PHR	Process Hazard Review, A systematic approach for identifying and assessing Process Hazards and risks and their possible effects.
PSSR	Pre-Startup Safety Review, A final review that must take place prior to "start-up" of the process affected by the Change to verify that relevant modifications have been thoroughly checked for safety, operability and maintainability prior to startup.

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APPENDIX

APPENDIX A - REVISION HISTORY LOG

TABLE 6. REVISION HISTORY LOG

Revision Number: 01

Date: 21/09/2023

Item Revised	Revision Description	Page No.
All	Document simplified as part of QMR initiative.	All
Remarks:		

APPENDIX B - LIST OF ABBREVIATIONS

TABLE 7. LIST OF ABBREVIATIONS

	'WHAT-IF' ANALYSIS GUIDELINE DOC NO: IS-HSE-GDL-316 REV. 01
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Abbreviation	Description
CME	Critical Mitigation Element
ESDV	Emergency Shut Down Valve
FTI	Facilities Technical Information
HAZOP	Hazard and Operability Study
IPL	Independent Protection Layers
KME	Key Mitigation Element
LOPA	Layers of Protection Analysis
LOPC	Loss of Primary Containment
LTI	Long Term Inhibit/ Isolation
MOC	Management of Change
NRO	Non-Routine Operations
PEC	Potentially Elevated Consequences
PFD	Process Flow Diagram
P&ID	Piping and Instrumentation Diagram
PHR	Process Hazard Review
PII	Permanent Incapacitating Injury
PSSR	Pre-Startup Safety Review

APPENDIX C - CONDITIONING FACTORS TYPE

Type	Description
General Condition	General Condition of the node
No Known Issues	No Issues
Deferral: AI-MI (Mechanical Integrity) AI-Mtce (Equipment Maintenance)	Inspection etc. (Fabric / Static equipment) Equipment Maintenance; Proof-test etc.

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Type	Description
NRO (Non-Routine Operation)	Non-Routine Operation in the node or affecting the node function
Organisational Change	Organisational Change
Procedural Change	Procedural Change
RIK (Replacement-in-Kind)	Replacement in Kind
Temporary Repair	Wraps; Clamps; other- Technical Reviews
LTI (Long-Term Isolation)	Long Term Inhibits- ESD, F&G, RV; Firewater
Related MOC's / MOC Requests	Related MOC's - planned or in-hand that may impact the review
Process / Maintenance Overrides	Planned short-term overrides
Historical Incidents	What? When? How? Why? Lessons Learned / Corrective actions?
SIL assessed	Have all the SIF (Safety Instrumented Functions) loops been SIL assessed and verified in the node?
No FTI	Is all key Facilities Technical Information available and as-built for the node?
PHA-Actions Outstanding	PHA Actions which have not yet been implemented: FEC; OPS; Especially JSA-transient works
Punch List Outstanding	Construction e.g. Temporary (wrong-specification material) bolts fitted; Cat. A; Cat. B
Procedural Issue	Procedure (NI; NC). Operations - chronic non-conformance or Procedure that does not exist or needs improvement
CAR (Corrective Action Request)	Audit; Incident - Various depts. All these - once accepted, if not actioned, constitute a subtle degradation of MOC
No FTI	e.g. Weather; Stress; Competencies
Deficiency: AI-MI AI-Maint.	< MAWT; Leak; Structural Damage Equipment / System Performance; Poor condition; Blocked RV / Impulse Lines;

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Type	Description
	Difficult to maintain; 'Ex' eqpmt. condition
Operating Phase	Change from Normal Ops., e.g. Start-up, shutdown, Emergency shutdown; Black start
Coincident Works	Simops; Triops
Defect Design	BV or Instrument access; Escape Routes compromised; Dead-legs; Risers External to jacket
Personnel in Area	Personnel in Area



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APPENDIX D - SAMPLE 'WHAT-IF' ANALYSIS WORKSHEET

System: 1. Ensure Clear definition of System limits
Sub-system: 1. Ensure Clear definition of Sub-System limits

Conditioning Factors
Type: Wraps, Long Term Inhibits, Deferrals, etc.

Automatically populated from the 'System' and 'Sub-system' sheet

If C>3 then the CME and supporting KME are to be categorised as 'HSE Critical'

In the event that no Additional Controls are identified then the Residual Risk (RR) shall be recorded at the same value as the Initial Risk (IR)

What-If	Hazard Scenario and Consequence	CAT	Comments	Intrinsic Controls : CMEs / KMEs	CAT	CME Name	HSE Crit Eqmnt?	Initial Risk			PEC Code	Additional Controls / Recommendations	CAT	CME Name	Comment	Responsibility	HSE Crit. Eqmnt?	Residual Risk		
								C	P	R								C	P	R
Provide Clear and concise description of Cause of Deviation	Provide clear and concise description of the Hazard Scenario and its potential consequences	Select Relevant Category from the drop down menu PAF/ER/PD/LOR\$/NOP	Document any significant comments which provide additional clarification of the Hazard Scenario and potential Consequences	Detail Intrinsic control relevant to the event scenario	Select Relevant Category from the drop down menu (Ref. Figure 7.7.1 above)	Select Relevant CME/KME Name from the drop down menu	If the consequence level is a 4 or 5 then a 'Y' shall be recorded in the column				Provide as necessary any comment clarifying why a CME may not be credited or Why the CME approach has not been used	Apply appropriate 'PEC Code' i.e. PEC1 PEC2 or PEC N/A (Ref. Figure 7.7.2 above)	Identify any robust additional controls or recommendations that the team identified	Select Relevant Category from the drop down menu (Ref. Figure 7.7.3 above)	Select Relevant CME/KME Name from the drop down menu	Add comment where necessary to support or justify implementation of A/C or Rec	Identify responsible party i.e. FETL/FOTL/ HSE Manager			
		PAF/ER/PD/LOR\$/NOP	Complete for each relevant CAT	Complete for each relevant CAT																
		PAF/ER/PD/LOR\$/NOP	Complete for each relevant CAT	Complete for each relevant CAT																
		PAF/ER/PD/LOR\$/NOP	Complete for each relevant CAT	Complete for each relevant CAT																